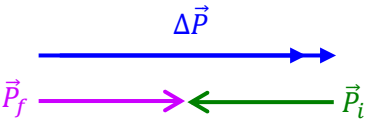


5	A	Magnitude of the change in momentum $\Delta P = \text{area under graph} = \frac{1}{2} (150)(40 \times 10^{-3}) = 3 \text{ Ns}$. This change in momentum takes place in the direction of the force, i.e. opposite to the initial momentum. In the vector diagram below, the ball is approaching the racket from right to left. $\vec{P}_f - \vec{P}_i = \Delta \vec{P}$ $P_f + P_i = \Delta P$ $(0.060)(30) + P_i = 3$ $P_i = 1.2 \text{ Ns}$ 
6	C	The thrust must provide an upward force to propel the rocket upward. This force must be least at equal to its